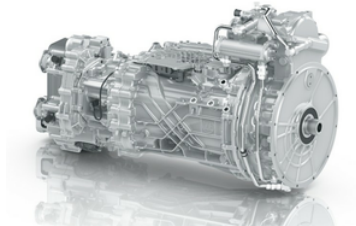
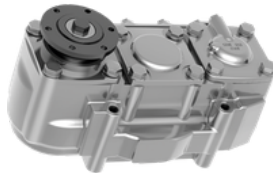
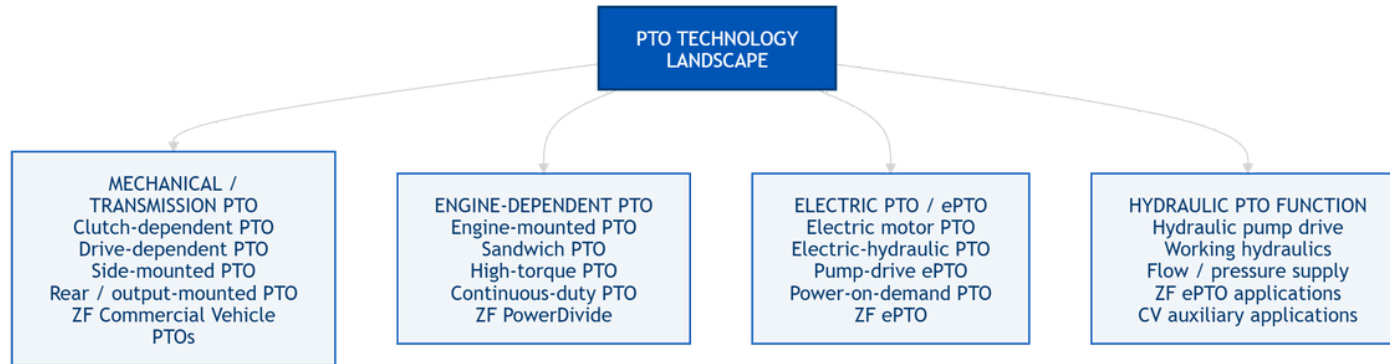


Power Take-Off Unit

Existing Technology, ZF Patent Landscape, and
Hydraulic-Side Patent Direction

It is the power bridge between vehicle drivetrain and external hydraulic work.



ZF Patent — US12169017B2

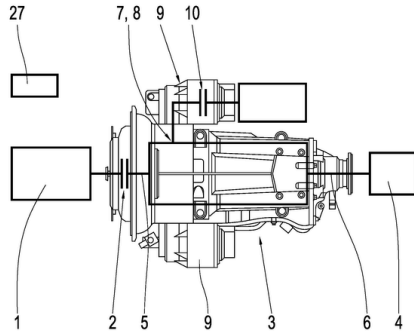


Fig. 1

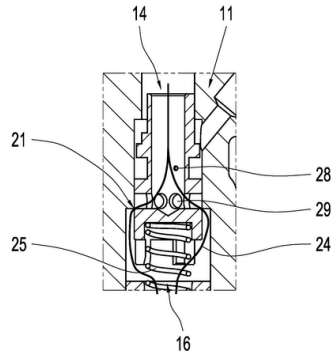


Fig. 4

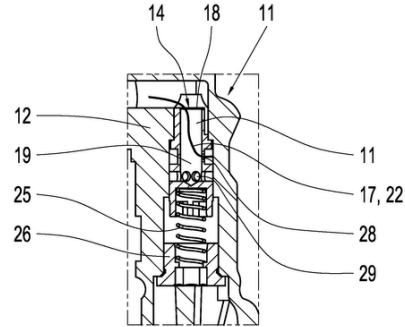


Fig. 2

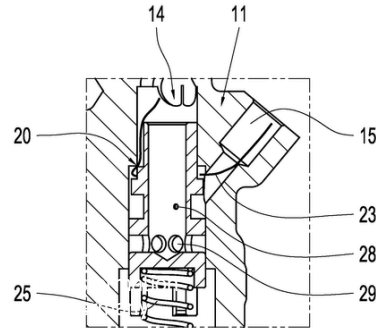


Fig. 3

Title: Vehicle transmission with a power take-off, vehicle and working machine arrangement

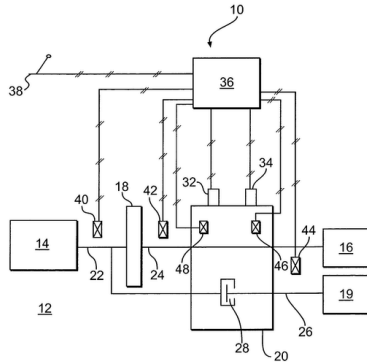
Priority: German application DE 10 2022 209 747.5

Assignee: ZF Friedrichshafen AG

Claims a PTO transmission lubrication system with reservoir, supply lines, return line, lubricant tapping point, and pressure-switched valve with bypass duct for PTO module lubrication.

Relevant Patents

US20030029691A1 — PTO clutch control system



Claims a PTO clutch control method where a controller modulates the pressure-control-valve signal using a single predetermined clutch-pressure profile, allowing smooth PTO clutch engagement across varying implement inertia.

US20140360290A1 / US9290092B2 — Chain drive PTO for hydraulic pump

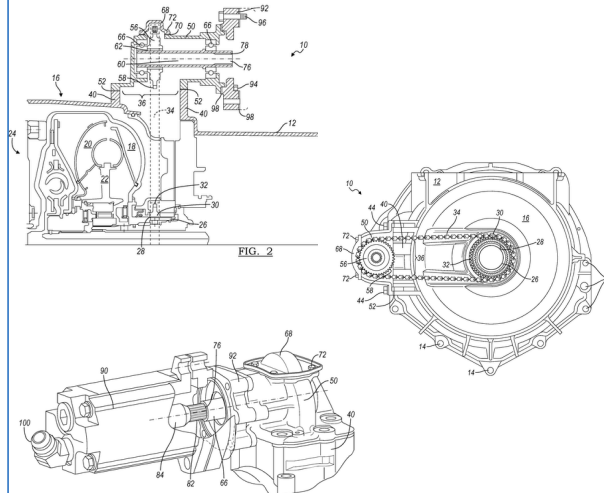
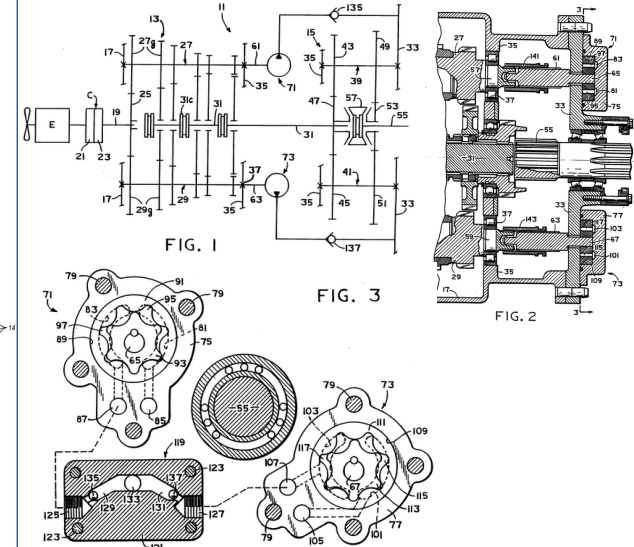


FIG. 3

Claims an off-axis hydraulic pump PTO where a torque-converter input member drives a first sprocket, a chain transfers torque to a second sprocket on a parallel offset stub shaft, and the stub shaft drives the hydraulic pump.

EP0631895A1 / US5492034A — Twin countershaft hydraulic PTO



Claims a PTO clutch control method where a controller modulates the pressure-control-valve signal using a single predetermined clutch-pressure profile, allowing smooth PTO clutch engagement across varying implement inertia.

Thank you.